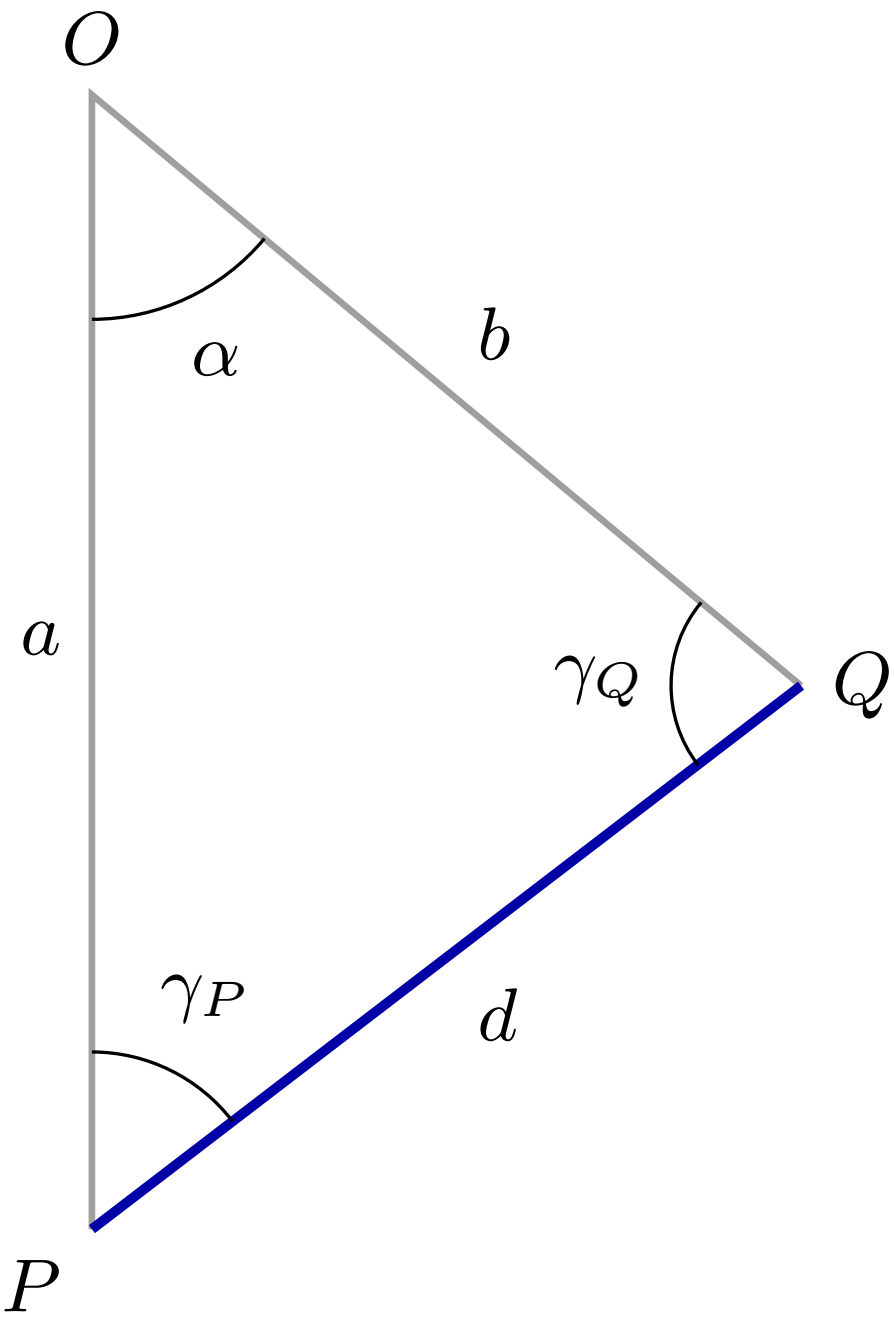

$$c_P = \frac{a - b\cos\alpha}{d}, \qquad c_Q = \frac{b - a\cos\alpha}{d}$$